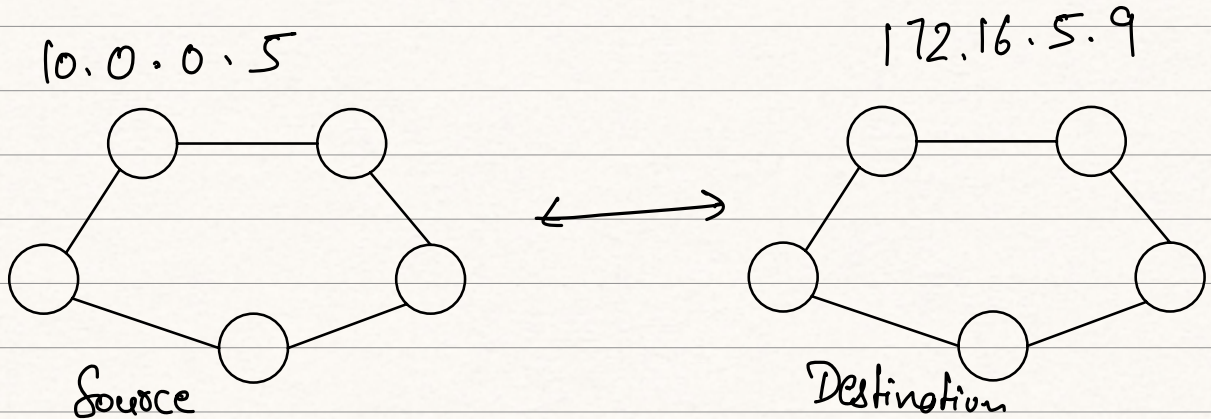


Subject: / /

- A (1-126)
- B (128-191)
- C (192-223)
- D (224-239)
- E (240-255)



for Broadcasting we use DBA

→ 172.16.255.255
DBA

* Communication b/w networks.

* UnICASTing

* BROADCASTing

SIP	DIP
10.0.0.5	172.16.5.9

SIP	DIP (DBA)
10.0.0.5	172.16.255.255

*

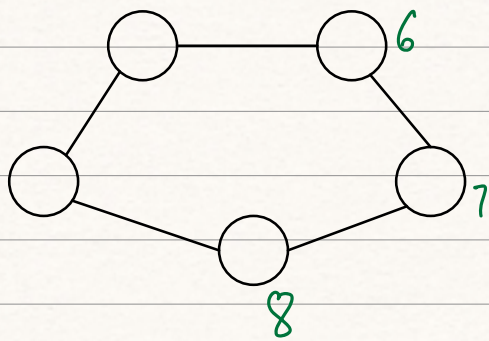
SIP	DIP
172.16.255.255	10.0.0.5

 , DBA always be use in DIP

DBA can't be in Source IP, so this is X.
only.

SIP	DIP
10.0.0.5	

Subject: * Communication within the network



SIP	DIP
10.0.0.5	10.0.0.7

* Special Case:-

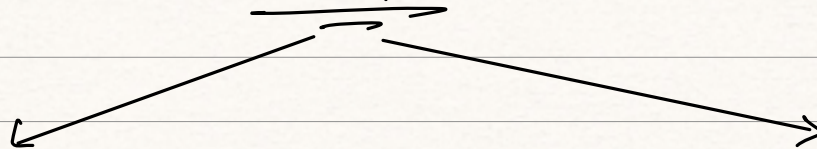
10.0.0.1	255.255.255.255
----------	-----------------

→ Scope in local (LAN)

* Within the lan for broadcasting 255.255.255.255, it works within the lan

* LBA will always be used as destination address
Local Broadcasting Address.

IP Address



Private

→ Scope is local

→ works only in LAN
→ free of cost

Public

→ Globally unique.

→ Under the control of
ISP (Internet Service
Provider)

→ not free of cost

→ Its range is:-

Subject:

10.0.0.0 - 10.255.255.255

172.16.0.0 - 172.31.255.255

192.168.0.0 - 192.168.255.255

→ In every LAN, we can use public IP addresses.

→ Drawbacks in classful:-

1. Class A $(2^7 - 2)N$

→ 80,00,000

each net $(2^{24} - 2)$ Host

2. Class B $(2^{14})N$

→ 655,34

each net $(2^{16} - 2)$ Host

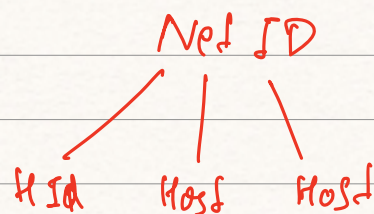
3. Class C $(2^{21})N$

→ 254

each net $(2^8 - 2)$ Host

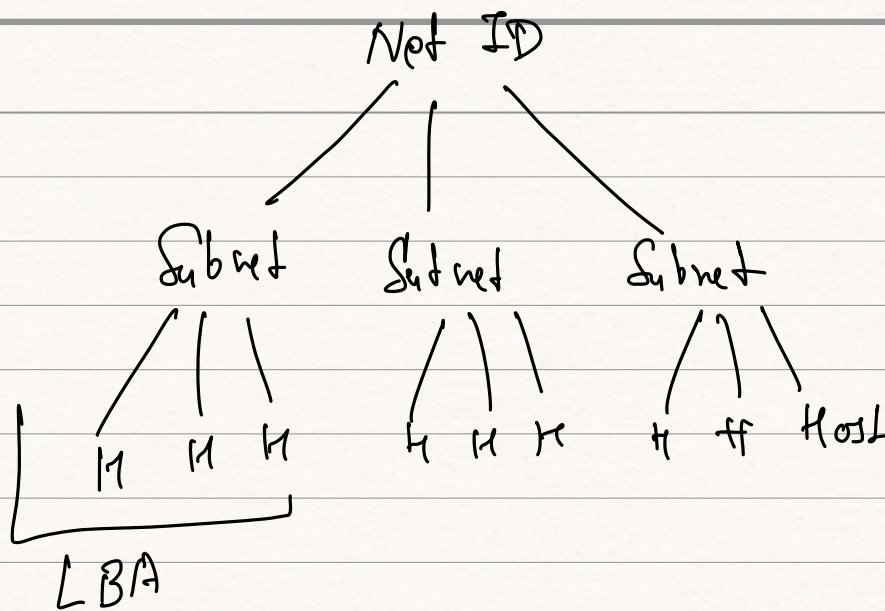
Drawback :- A lot of IP address in class A and B, wastages of IP address occurs.

2-level hierarchy



Subnetting :- Dividing the net. in small parts for effective utilization of IP address is known as subnetting

Subject: / /



1. In class C, if subnet mask is

255.255.255.224, no. of subnets = ?

⇒

By default Class C Mask,

no. of host in each subnet = ?

(255.255.255.0)

11111111.11111111.11111111.00000000
N

111 00000
↓ ↓
Subnet Host

No. of subnets = $2^3 - 2$ Subnet ID
LBA

No. of host in each subnet = $2^5 - 2$

* During subnetting, subnet bits are borrowed from the host position

Subject: / /

2. In class C, if subnet mask is

255.255.255.240

$\underbrace{11111}_{\text{Subnet}} \underbrace{000}_{\text{Host}}$
 $2^5 - 2$ $2^3 - 2$

2	240	0
2	120	0
2	60	0
2	30	1
2	15	1
2	7	1
2	3	1
	1	

3. In class of subnet mask is 255.255.254.0.

2	254	0
2	127	1
2	63	1
2	31	1
2	15	1
2	7	1
2	3	1
	1	

$\underbrace{1111111}_{} \underbrace{0}_{\text{Host}} \cdot \underbrace{00000000}_{\text{Host}}$
 $2^7 - 2$ $2^8 - 2$

Q. IP = 201.55.79.99

Subnet Mask = 255.255.255.224

Subnet ID :

Subnet No.1.

Subject: $\Rightarrow$

99 $\rightarrow$

//

01100011

224 $\rightarrow$

11100000

01100000

$\downarrow \quad \downarrow$
 $2^6 \quad 2^5$

$$64 + 32 = \underline{96}$$

Subnet ID = 201.55.79.96

Subnet no.: 011
 $\hookrightarrow$ 3

2	99	1
2	49	1
2	24	0
2	12	0
2	6	0
2	3	1
1		